

Homework Assignment 9

1. Consider operation of an ideal thrust chamber operating at 11 km altitude ($p_a = 22.7$ kPa). It has the following characteristics:

- Stagnation pressure 1 MPa
- Stagnation temperature 3200 K
- Throat area 0.035 m^2
- Exit area 0.875 m^2
- Propellant gas has $\gamma = 1.25$ and molar mass ($\bar{M}$) 12

(a) Verify that the exit velocity of the flow is supersonic. {Ans. $\left(\frac{p_0}{p_a}\right)_{crit} = 16.45$ }

(b) Calculate the exit conditions (Mach number, velocity, pressure) of the jet. {Ans.: 4.15, 3889 m/s, 3.19 kPa}

(c) Determine the mass flow rate and the thrust generated {Ans.: 15.45 kg/s, 43.01 kN}

(d) From the results of part (c), calculate the characteristic velocity and the thrust coefficient. {Ans.: 2265 m/s, 1.229}

2. If the thrust chamber of Problem 1 were operated at sea level ($p_a = 101$ kPa), would the exit jet be supersonic? Explain. If it is not, where will the shock be located: at the exit plane? Inside the nozzle? Again, explain.

3. Consider a rocket with thrust coefficient $C_\tau = 1.50$. The exhaust jet is overexpanded. Calculate the area ratio (A_e/A_t) and the exit Mach number (M_e). The values calculated must yield the thrust coefficient correct to three significant figures; that is, to two decimal places in this case. $p_a/p_0 = 0.01$. $\gamma = 1.23$.